

Optimization of GIDL Assisted Erase Operation in 3D NAND Flash Memory

*BTP Thesis submitted to the
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by

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CERTIFICATE

This is to certify that the thesis entitled **Optimization of GIDL Assisted Erase Operation in 3D NAND Flash Memory**, submitted by **Pulla Abhivamsh** to Indian Institute of Technology Bhubaneswar, is a record of bonafide research work under my supervision and I consider it worthy of consideration for the award of the degree of Bachelor of Technology of the Institute.

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Abstract

3D NAND flash memory has become a pivotal technology in modern data storage systems due to its scalability and high-density capabilities. Optimizing its performance requires a deep understanding of device characteristics and mechanisms such as Gate Induced Drain Leakage (GIDL), which plays a critical role in various operations. This project investigates the dependence of GIDL on key parameters in a NAND memory string to provide insights for performance enhancement.

Initially, a NAND memory string was simulated based on a reference design from IEEE literature, followed by the simulation of its I_D - V_{GS} characteristics to validate the memory string's operational behavior. The study then focused on analyzing the GIDL phenomenon, exploring its dependence on parameters such as BL plug doping concentration, Gate metal selection, Trap charge concentration, and Fixed charge concentration, Oxide layer thickness. The simulations showed how these parameters affect GIDL behavior, providing important insights for improving device performance and reliability.

These findings lay the groundwork for further investigations into employing GIDL-assisted mechanisms, such as erase operations, to improve erase efficiency and reliability in 3D NAND Flash Memory.

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Chapter 1

Introduction

1.1 Background

NAND flash memory is a non-volatile memory which doesn't require a regular power supply to work and retain data. NAND memory cells are made up of two types of gates: control and floating gates. Both gates are used to regulate the flow of data. NAND is commonly found in memory cards, USB flash drives, and solid-state drives. NAND flash memory is evolving to meet the demands of modern data storage. Its robustness, speed, and cost-effectiveness make it a key player in the ever-changing world of memory storage technology.

1.1.1 Erase Operation in NAND

Erase operation in a NAND cell can be performed by ejecting electrons from the floating gate of a cell in 2 ways - Drain/source Fowler–Nordheim(FN) ejection and a Channel FN ejection [1] [2]. In selected blocks, grounded wordlines and floating unselected ones, a 20 V erase pulse applied to the p-well creating a 20 V potential difference between the p-well and control gates. This induces FN tunneling of electrons from the floating gate to the p-well, shifting the cell threshold voltage negatively, effectively erasing the cell.

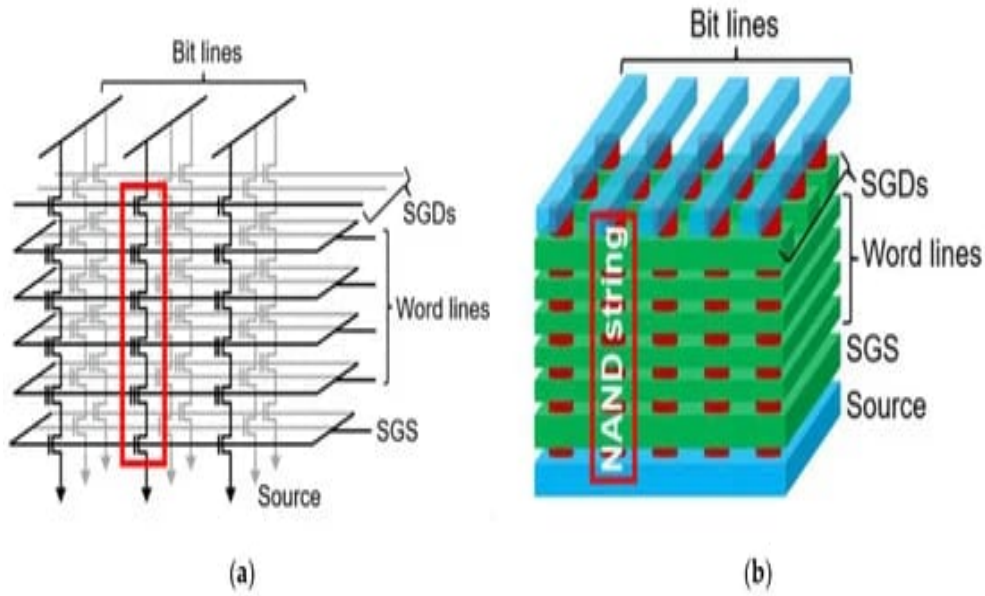


Figure 1.1: 3D NAND Array

1.2 Gate Induced Drain Leakage(GIDL)

GIDL is a leakage current that occurs in the drain region of a MOSFET when a high gate-to-drain electric field exists, typically under conditions of a negative gate voltage and a high drain voltage. Current is produced mainly by minority charge carriers. This phenomenon is particularly relevant in NAND flash memory due to the unique operating conditions of these devices. NAND flash memory cells utilize floating-gate or charge-trap structures to store data. During certain operations, such as the erase operation, high fields in the drain region can cause GIDL [3].

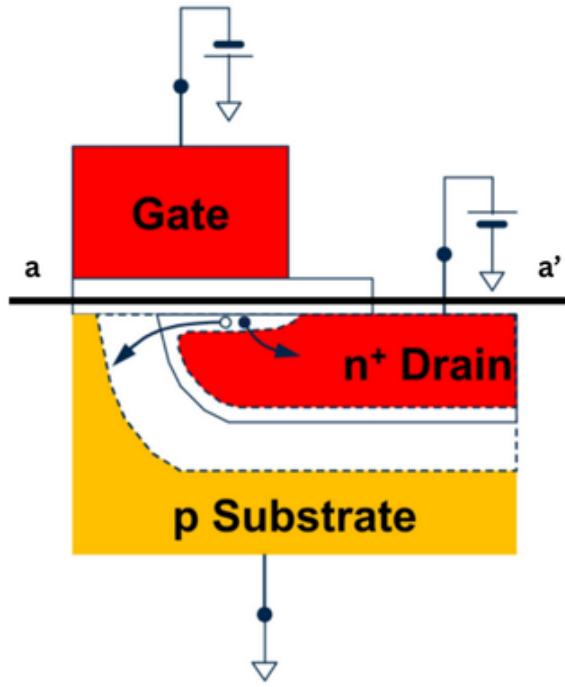


Figure 1.2: Illustration of GIDL mechanism

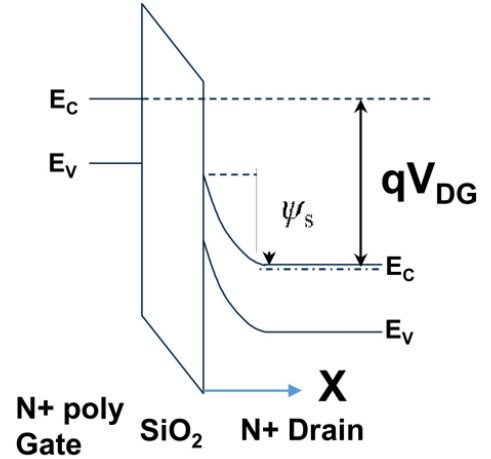


Figure 1.3: Energy band diagram along aa' line

Several parameters can affect the magnitude of GIDL in NAND flash devices:

1. Doping concentration of BL plug : The doping concentration of the bitline (BL) plug determines the electric field intensity near the drain junction. A higher doping concentration creates a steeper electric field gradient at the drain junction.
2. Gate material : The work function of the gate material influences the threshold voltage and the electric potential in the gate-drain overlap region.
3. Trap charge and fixed charge concentrations : Trap and fixed charges in the oxide layer or at the oxide-semiconductor interface modulate the local electric field and potential distribution.
4. Voltage biasing : GIDL is strongly dependent on the gate-to-drain and drain-to-source voltage configurations.
5. Temperature - Temperature influences the bandgap energy of the semiconductor, carrier mobility, and BTBT rates.
6. Oxide layer thickness - A high vertical electric field between the gate and the drain during

GIDL. The thickness of the gate oxide directly influences the strength of this electric field.

1.2.1 GIDL Assisted Erase Operation

During the erase operation, electrons are removed from the floating gate by tunneling through the oxide layer into the channel or substrate. High electric fields applied to the gate and drain regions enhance the GIDL current inducing minority carriers to drain, facilitating tunneling. This leakage current effectively reduces the threshold voltage of the memory cell, completing the erase operation. [4]

Advantages of GIDL-Assisted Erase

1. Low voltage: GIDL requires lower gate voltages compared to FN tunneling, reducing stress on the oxide.
2. Faster erase Speed: High electric fields near the junction enable rapid electron removal.
3. Improved endurance: Less oxide damage ensures longer device lifetime.

1.2.2 Impact of GIDL on Erase Efficiency and Reliability

Gate-Induced Drain Leakage (GIDL) influences the erase process in NAND flash memory by enhancing electron removal, improving erase speed. However, excessive GIDL effects impact cell endurance and retention. Optimizing GIDL through careful control of doping concentration, gate material, and oxide properties is crucial to achieving a balance between erase efficiency and long-term reliability in NAND memory devices.

Chapter 2

Simulator Calibration

2.1 3D NAND String Structure

This 3D NAND string structure, derived from the paper [5] features a total of 48 word lines (WLs) and includes two dummy cells, select gates, and a bitline contact. Key specifications include a silicon channel radius of 15 nm, a silicon thickness of 10 nm, and a multi-layer ONO stack. The doping concentrations for the bitline (N_D^{BL}) and the source line (N_D^{SL}) regions are 10^{19} cm^{-3} . For our study, we have reduced the number of word lines to 8 in the NAND string.

Table 2.1: Parameters of NAND string used in this study [5]

L_{WL}	28 nm
L_S	28 nm
t_{si}	10 nm
r_t	15 nm
O/N/O	5 / 5 / 6 nm
$N_D^{BL} = N_D^{SL}$	10^{19} cm^{-3}
N_D^B	10^{15} cm^{-3}
WLs	12

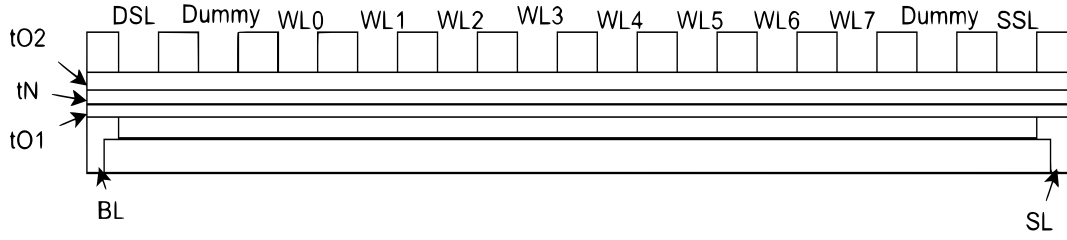


Figure 2.1: Cross-sectional schematic of NAND string used in this work

2.2 Methodology

2.2.1 Calibration of I_D - V_{GS} Curve

The initial step involved calibrating the I_D - V_{GS} curve of the simulated structure using the reference research paper [6]. This ensured that the simulated results aligned with the experimental data reported in the paper by tuning parameters such as mobility, fixed and trapped charge concentrations at interface, establishing a baseline for further analysis.

2.2.2 Study of GIDL Parameter Effects

Following calibration, the impact of various GIDL-related parameters on hole density within the channel was investigated. The GIDL parameters studied are

- 1) Fixed charge and trap charge concentration : Hole current and Hole Density profiles are observed for varied concentration values
- 2) Gate metal variation : Workfunctions of metal contacts are varied to observe the effect of different metals on GIDL current and Hole density.
- 3) BL plug doping variation : BL plug doping concentration is varied.
- 4) Oxide layer thickness : Thickness of tunneling oxide layer(tO_1) of the NAND string is varied.

In this study, the behavior of the hole current in the channel was analyzed under varying potential differences between the gate and drain. The insights gained from this analysis provide a deeper understanding of how gate-drain voltage differences influence the behavior of

hole transport and the generation of Gate-Induced Drain Leakage (GIDL) current in NAND structures.

The paper [7] elaborates on the critical role that these voltage variations play in modulating the hole current. Specifically, the potential difference between the gate and drain creates a localized electric field that affects the hole injection process. When the drain voltage is increased relative to the gate, the electric field at the drain junction becomes more pronounced, enhancing the tunneling of electrons and holes, which contributes significantly to GIDL current.

Chapter 3

Results and Discussions

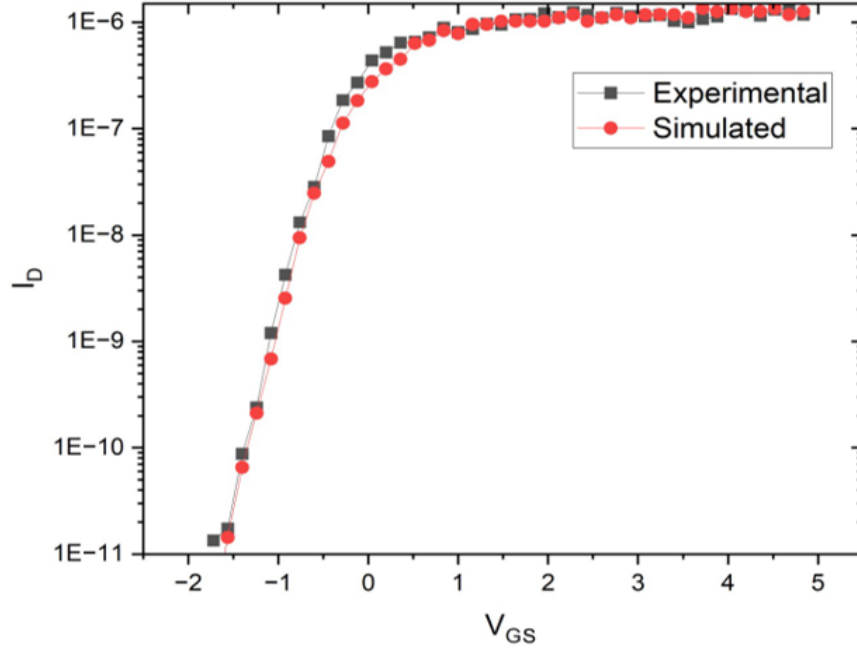
3.1 3D NAND String Calibration

In NAND string memory calibration, mobility and fixed charge and trap charge concentrations of interface were tuned to match the I_D - V_{GS} plot[Fig.3.1], ensuring accurate simulation of carrier transport and threshold voltage effects.

3.2 Analysis of GIDL Parameters

This section presents the analysis of key parameters Hole current and Hole density, affecting Gate Induced Drain Leakage, including drain voltage, gate metal, BL plug doping concentration, and Trap and Fixed charge concentrations and oxide layer thickness. These parameters were systematically varied to evaluate their influence on GIDL behavior and overall device performance. The results, summarized, offer insights into how each parameter contributes to GIDL variation, helping to identify critical factors for optimizing erase operations in 3D NAND flash memory. The findings provide a deeper understanding of GIDL mechanisms for improved device design.

The following are inferred from the plots obtained

Figure 3.1: Calibrated I_D - V_{GS} plot

3.2.1 Dependency of GIDL on BTBT Parameter

We can observe more hole density with BTBT model included, which shows the dependency of GIDL on BTBT parameter [Fig. 3.2]. BTBT is a quantum mechanical process in which electrons tunnel directly from the valence band of the substrate to the conduction band of the drain, generating electron-hole pairs that result in leakage current even when the transistor is off. The magnitude of this BTBT-generated current—and hence GIDL—is governed by specific parameters in the BTBT generation model.

3.2.2 Dependency on Fixed and Trap Charge Concentration

We can observe that as we increase the fixed charge, hole density is acquiring a linear profile across the channel [Fig. 3.4]. For lower fixed charge concentrations, hole density is more near the drain-gate junction and decreasing linearly across the channel.

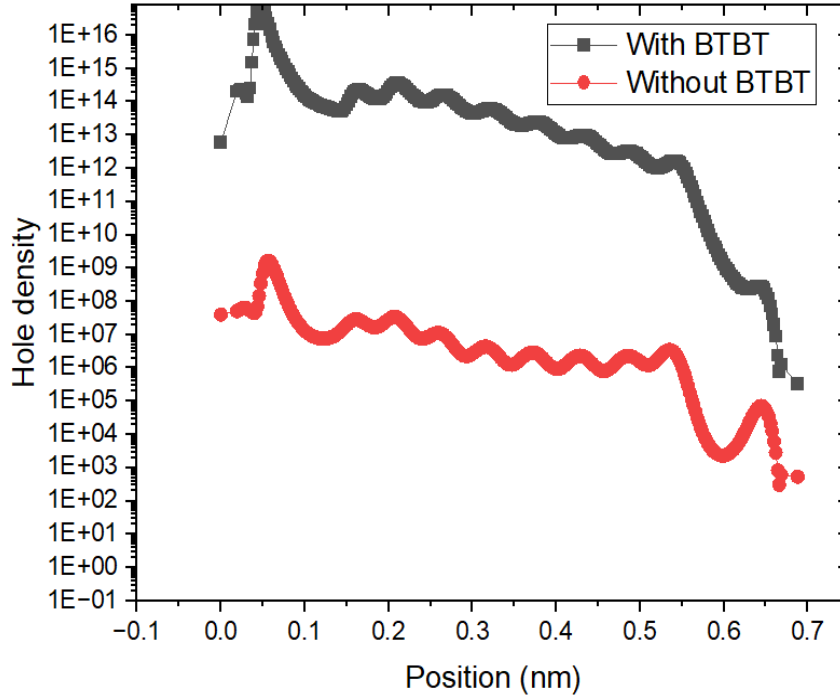


Figure 3.2: Hole density distribution with and without BTBT model included in simulation

3.2.3 Dependency on Gate Metal

As V_{BL} increases, we can observe that the hole current starts increasing earlier for metal with higher work function when compared with lower work function metals[Fig.3.3]. Hole density is also higher for metals with higher work function and it gradually decreases as work function decreases.

3.2.4 Dependency on BL Plug Doping Concentration

The Gate-Induced Drain Leakage (GIDL) current is strongly influenced by the doping concentration of the bit-line (BL) plug region, which is typically an n+ drain extension in a NAND string. This region lies underneath the gate-to-drain overlap and plays a critical role in determining the electric field profile at the tunneling junction. A higher BL plug doping concentration increases the abruptness of the drain-to-channel junction, leading to a steeper band bending and stronger electric field at the silicon surface when a high drain voltage and negative gate voltage are applied. This enhanced electric field significantly increases the Band-to-Band Tunneling (BTBT) rate, thereby elevating the GIDL current and hole density in the channel[Fig.3.5].

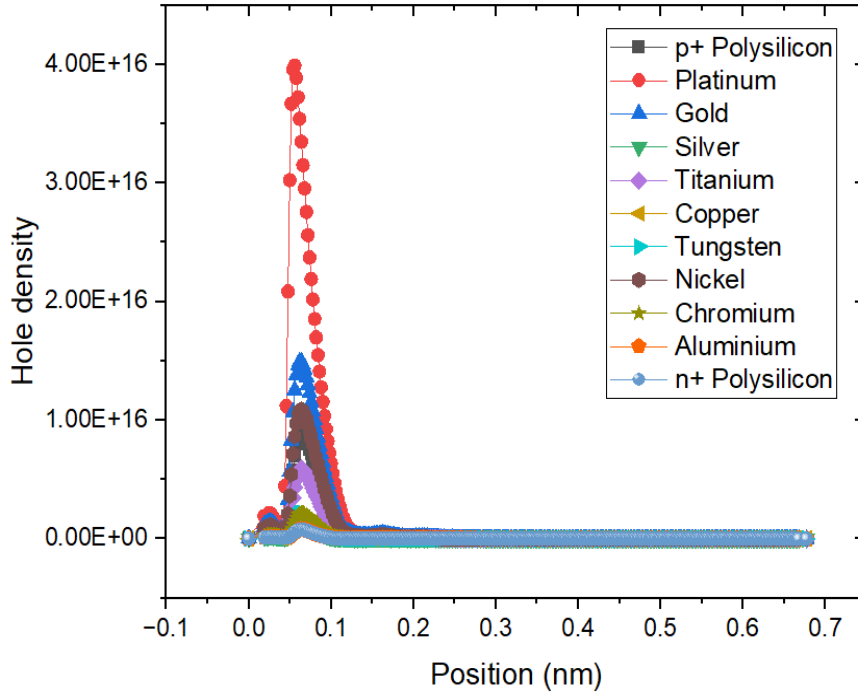
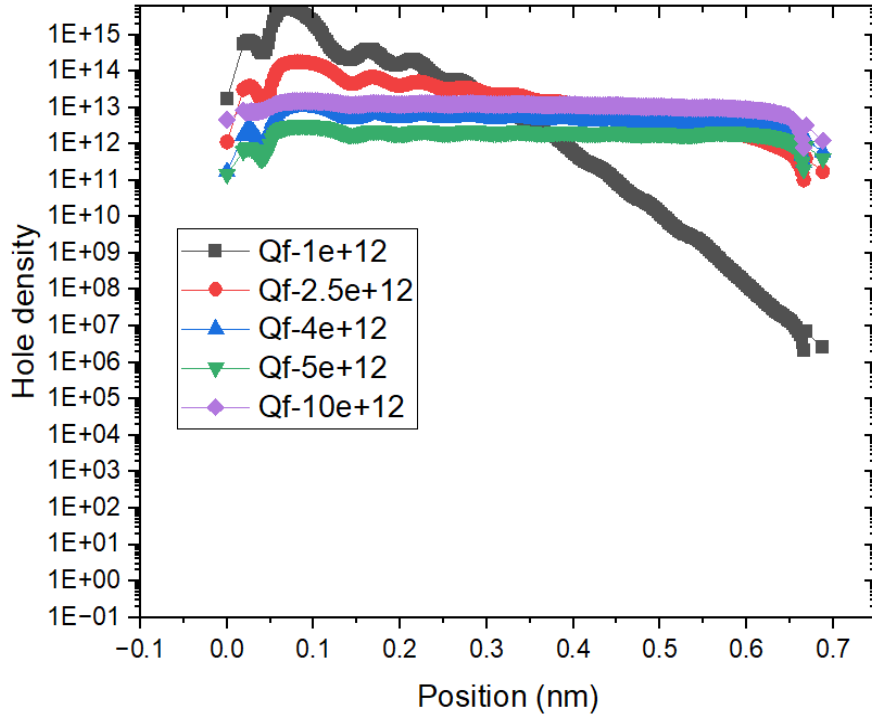


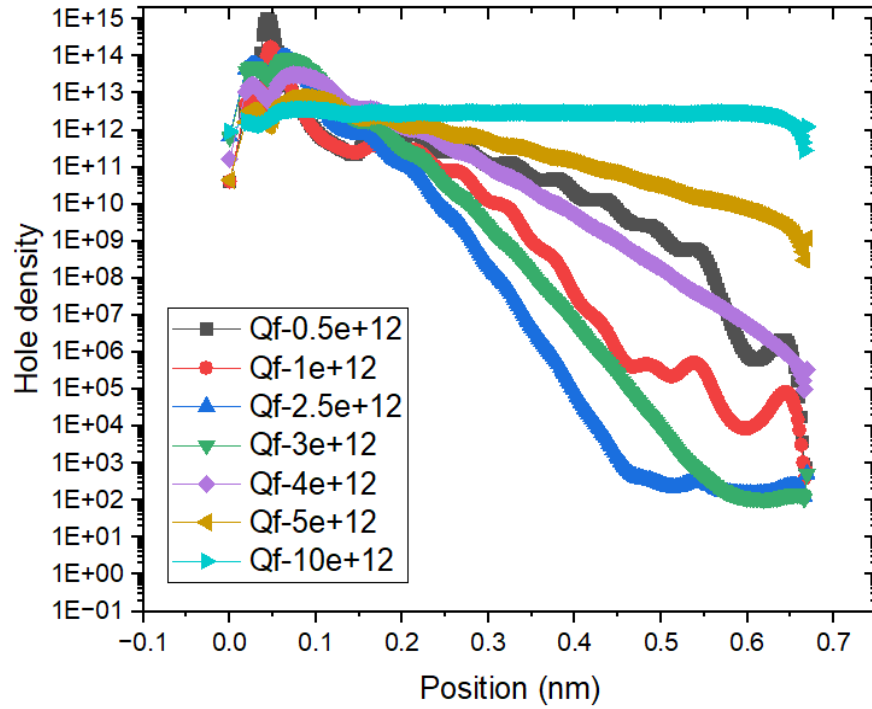
Figure 3.3: Hole density for different gate metals

3.2.5 Dependency on Oxide Layer Thickness

Gate-Induced Drain Leakage (GIDL) is highly sensitive to the thickness of the gate oxide layer. This is because the oxide thickness directly affects the vertical electric field between the gate and the drain region, particularly near the drain edge where GIDL primarily occurs. When the oxide layer is thinner, the electric field for a given gate-to-drain voltage becomes stronger, which significantly enhances band-to-band tunneling (BTBT) between the valence band and conduction band in the drain region. As a result, thinner oxide layers lead to a substantial increase in GIDL current.



(a)



(b)

Figure 3.4: (a)Hole density distribution when trap concentration = $1 \times 10^{12} \text{ cm}^{-3}$ (b)Hole density distribution when trap concentration = $5 \times 10^{12} \text{ cm}^{-3}$

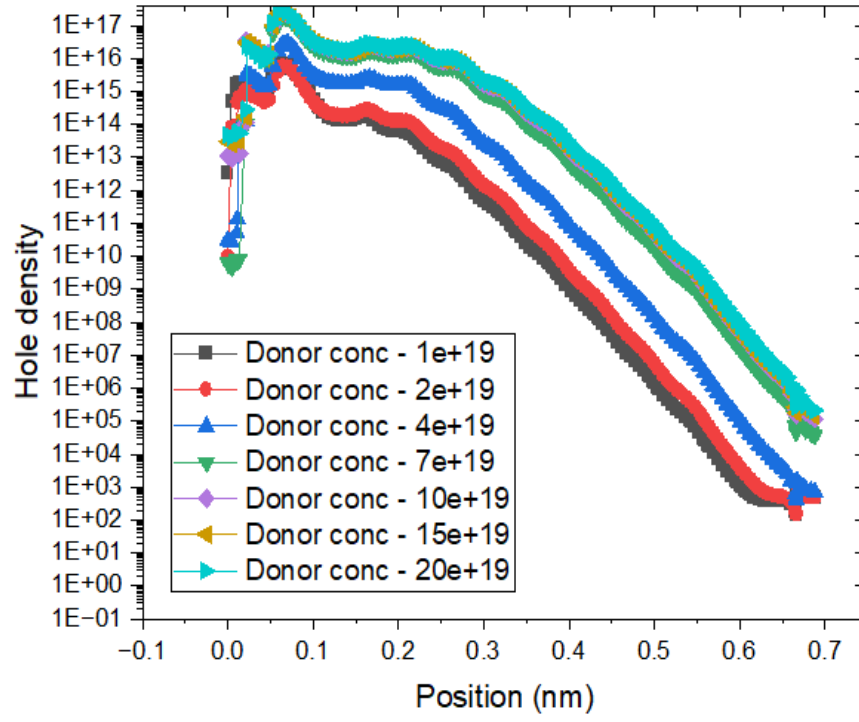


Figure 3.5: Hole density distribution for varying BL plug doping concentrations

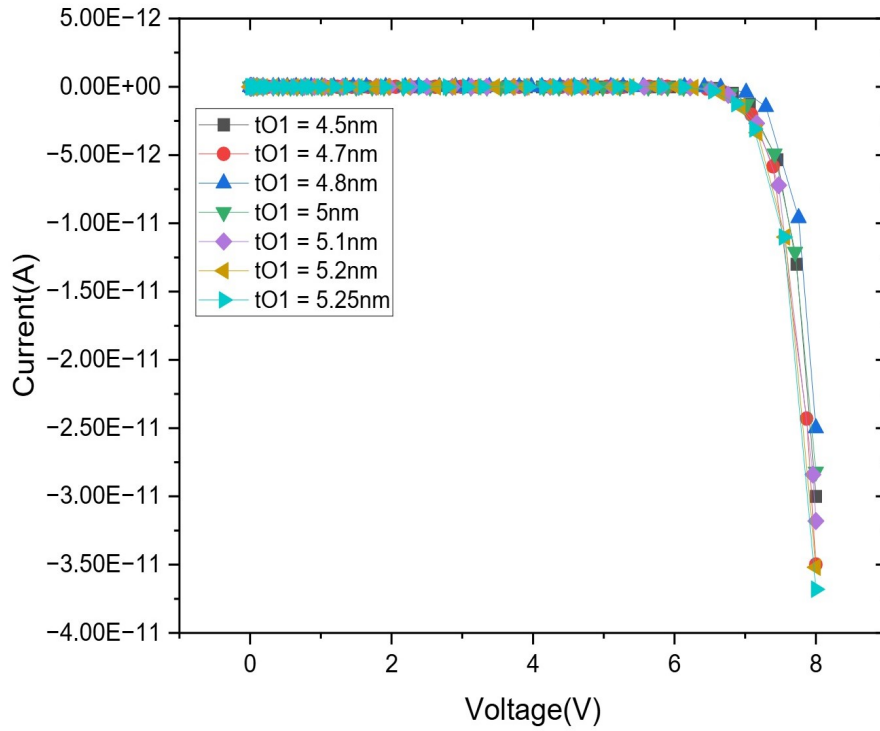


Figure 3.6: Hole current for varying thickness of tunneling oxide

Chapter 4

Future Scope

The current simulations have been conducted using a crystalline silicon channel, future work will focus on extending the study to polycrystalline silicon (poly-Si) channels. This transition is crucial for enhancing the practicality and real-world relevance of the results, as poly-Si is commonly used in actual 3D NAND flash memory fabrication. Future simulations will involve analyzing the impact of grain boundaries and grain-assisted trap states inherent to poly-Si, and studying their influence on GIDL behavior. This will include varying grain size, grain boundary distribution, and associated trap densities to better understand their effect on critical device parameters and optimize erase operations under realistic structural conditions.

Additionally, future studies will aim to analyze the optimal operating conditions for the erase operation of the NAND string. By simulating different biasing schemes and structural variations, the goal will be to minimize GIDL-induced degradation while ensuring efficient GIDL-assisted erase functionality. This will contribute to a more robust understanding of device behavior under realistic manufacturing conditions and help in optimizing the reliability and performance of future NAND flash technologies.

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